

Finals Practice Task 1.2 Data Preparation and Exploratory Data Analysis (EDA) using Mysql

Description:

This task focused on using SQL and MySQL Workbench to clean, prepare, and analyze a call center dataset. The process started by creating a database schema and designing a table structure that matched the CSV dataset. The dataset was then imported into MySQL using the Table Data Import Wizard.

Several data cleaning techniques were applied to improve data accuracy and consistency. The `call_timestamp` column was converted into the proper MySQL date format, while incorrect values in the `csat_score` column were handled by replacing invalid zero values with NULL values. Different SQL update queries were also used to fix and standardize the dataset.

After cleaning the data, exploratory data analysis (EDA) was performed using SQL queries. Different aggregations and calculations were created to analyze call center performance, including identifying the busiest days, checking service-level agreement performance, and calculating minimum, maximum, and average call durations. Window functions and grouping queries were also used to gain additional insights from the dataset.

This activity strengthened skills in SQL querying, database management, data cleaning, and data analysis using MySQL Workbench.

